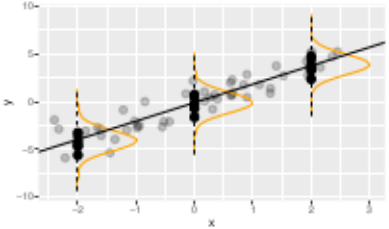
MEASURING DISTRIBUTION SIMILARITY IN ML / 2

- **Probabilistic model fitting**
  Assume our learner is probabilistic, i.e., we model $p(y|\mathbf{x})$ (for example, logistic regression, Gaussian process, ...).

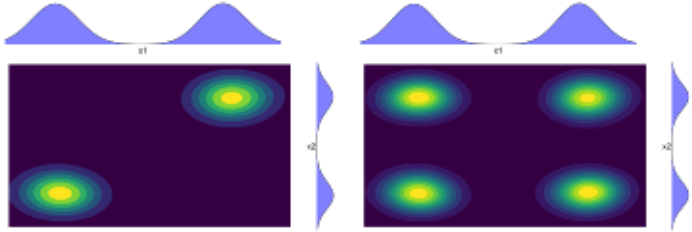


We want to minimize the difference between $p(y|\mathbf{x})$ and the conditional data generating process $\mathbb{P}_{y|\mathbf{x}}$ based on the data stemming from $\mathbb{P}_{y,\mathbf{x}}$.

Many losses can be derived this way. (e.g., cross-entropy loss)

MEASURING DISTRIBUTION SIMILARITY IN ML / 3

- **Feature selection** In feature selection, we want to choose features the target strongly depends on.



We can measure dependency by measuring the similarity between $p(\mathbf{x}, y)$ and $p(\mathbf{x}) \cdot p(y)$.

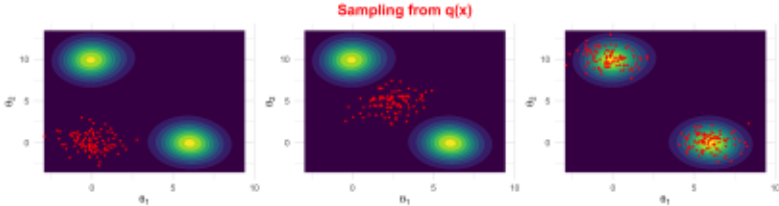
We will later see that measuring this similarity with KL leads to the concept of mutual information.

MEASURING DISTRIBUTION SIMILARITY IN ML / 4

- **Variational inference (VI)** By Bayes' theorem it holds that the posterior density

$$p(\theta|\mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{X}, \theta)p(\theta)}{\int p(\mathbf{y}|\mathbf{X}, \theta)p(\theta)d\theta}.$$

However, computing the normaliziation constant $c = \int p(\mathbf{y}|\mathbf{X}, \theta)p(\theta)d\theta$ analytically is usually intractable.



In VI, we want to fit a density q_ϕ with parameters ϕ to $p(\theta|\mathbf{X}, \mathbf{y})$.

KL DIVERGENCE / 2

- **Forward KL for probabilistic model fitting**
  We have samples from the DGP $p(y, \mathbf{x})$ when we fit our ML model.

  If we have a probabilistic ML model q_ϕ the expected forward KL

$$\mathrm{E}_{\mathbf{x}\sim p_\mathbf{x}} D_{KL}(p(\cdot|\mathbf{x})\|q_\phi(\cdot|\mathbf{x})) = \mathrm{E}_{\mathbf{x}\sim p_\mathbf{x}} \mathrm{E}_{y\sim p_{y|\mathbf{x}}} \log\left(\frac{p(y|\mathbf{x})}{q_\phi(y|\mathbf{x})}\right).$$

  We can directly minimize this objective since

$$\begin{aligned}\nabla_\phi \mathrm{E}_{\mathbf{x}\sim p_\mathbf{x}} D_{KL}(p(\cdot|\mathbf{x})\|q_\phi(\cdot|\mathbf{x})) &= \mathrm{E}_{\mathbf{x}\sim p_\mathbf{x}} \mathrm{E}_{y\sim p_{y|\mathbf{x}}} \nabla_\phi \log\left(p(y|\mathbf{x})\right) \\ &\quad - \mathrm{E}_{\mathbf{x}\sim p_\mathbf{x}} \mathrm{E}_{y\sim p_{y|\mathbf{x}}} \nabla_\phi \log\left(q_\phi(y|\mathbf{x})\right) \\ &= -\nabla_\phi \mathrm{E}_{\mathbf{x}\sim p_\mathbf{x}} \mathrm{E}_{y\sim p_{y|\mathbf{x}}} \log\left(q_\phi(y|\mathbf{x})\right)\end{aligned}$$

  $\Rightarrow$ We can estimate the gradient of the expected forward KL without bias, although we can not evaluate $p(y|\mathbf{x})$ in general.

KL DIVERGENCE / 3

- **Reverse KL for VI**
  Here, we know our target density $p(\theta|\mathbf{X}, \mathbf{y})$ only up to the normalization constant, and we do not have samples from it.

  We can directly apply the reverse KL since for any $c \in \mathbb{R}_+$

$$\begin{aligned}\nabla_\phi D_{KL}(q_\phi \| p) &= \nabla_\phi \mathrm{E}_{\theta \sim q_\phi} \log\left(\frac{q_\phi(\theta)}{p(\theta)}\right) \\ &= \nabla_\phi \mathrm{E}_{\theta \sim q_\phi} \log\left(\frac{q_\phi(\theta)}{p(\theta)}\right) - \underbrace{\nabla_\phi \mathrm{E}_{\theta \sim q_\phi} \log c}_{=0} \\ &= \nabla_\phi \mathrm{E}_{\theta \sim q_\phi} \log\left(\frac{q_\phi(\theta)}{c \cdot p(\theta)}\right).\end{aligned}$$

  $\Rightarrow$ We can estimate the gradient of the reverse KL without bias (even if we only have an unnormalized target distribution)

KL DIVERGENCE / 4

The asymmetry of the KL has the following implications

- Forward KL $D_{KL}(p\|q_\phi) = \mathbb{E}_{\mathbf{x}\sim p} \log\left(\frac{p(\mathbf{x})}{q_\phi(\mathbf{x})}\right)$ is mass-covering since $p(\mathbf{x}) \log\left(\frac{p(\mathbf{x})}{q_\phi(\mathbf{x})}\right) \approx 0$ if $p(\mathbf{x}) \approx 0$ and $q_\phi(\mathbf{x}) \not\gg p(\mathbf{x})$.
- Reverse KL $D_{KL}(q_\phi\|p) = \mathbb{E}_{\mathbf{x}\sim q_\phi} \log\left(\frac{q_\phi(\mathbf{x})}{p(\mathbf{x})}\right)$ is mode-seeking (zero-avoiding) since $q_\phi(\mathbf{x}) \log\left(\frac{q_\phi(\mathbf{x})}{p(\mathbf{x})}\right) \gg 0$ if $p(\mathbf{x}) \approx 0$ and $q_\phi(\mathbf{x}) > 0$

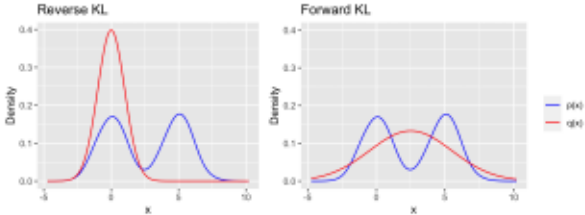


Figure: Optimal q_ϕ when q_ϕ is restricted to be Gaussian.